

[Omega-3 Fatty Acids: Fact Sheet for Consumers \(original version\)](#)

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What are omega-3 fatty acids and what do they do?

Omega-3 fatty acids are found in foods, such as fish and flaxseed, and in dietary supplements, such as fish oil.

The three main omega-3 fatty acids are alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA). ALA is found mainly in plant oils such as flaxseed, soybean, and canola oils. DHA and EPA are found in fish and other seafood.

ALA is an essential fatty acid, meaning that your body can't make it, so you must get it from the foods and beverages you consume. Your body can convert some ALA into EPA and then to DHA, but only in very small amounts. Therefore, getting EPA and DHA from foods (and dietary supplements if you take them) is the only practical way to increase levels of these omega-3 fatty acids in your body.

Omega-3s are important components of the membranes that surround each cell in your body. DHA levels are especially high in retina (eye), brain, and sperm cells. Omega-3s also provide calories to give your body energy and have many functions in your heart, blood vessels, lungs, immune system, and endocrine system (the network of hormone-producing glands).

How much omega-3s do I need?

Experts have not established recommended amounts for omega-3 fatty acids, except for ALA. Average daily recommended amounts for ALA are listed below in grams (g). The amount you need depends on your age and sex.

Life Stage	Recommended Amount of ALA
Birth to 12 months*	0.5 g
Children 1–3 years	0.7 g
Children 4–8 years	0.9 g
Boys 9–13 years	1.2 g
Girls 9–13 years	1.0 g
Teen boys 14–18 years	1.6 g
Teen girls 14–18 years	1.1 g

Life Stage	Recommended Amount of ALA
Men	1.6 g
Women	1.1 g
Pregnant teens and women	1.4 g
Breastfeeding teens and women	1.3 g

*As total omega-3s. All other values are for ALA alone.

What foods provide omega-3s?

Omega-3s are found naturally in some foods and are added to some fortified foods. You can get adequate amounts of omega-3s by eating a variety of foods, including the following:

- Fish and other seafood (especially cold-water fatty fish, such as salmon, mackerel, tuna, herring, and sardines)
- Nuts and seeds (such as flaxseed, chia seeds, and walnuts)
- Plant oils (such as flaxseed oil, soybean oil, and canola oil)
- Fortified foods (such as certain brands of eggs, yogurt, juices, milk, soy beverages, and infant formulas)

What kinds of omega-3 dietary supplements are available?

Omega-3 dietary supplements include fish oil, krill oil, cod liver oil, and algal oil (a vegetarian source that comes from algae). They provide a wide range of doses and forms of omega-3s.

Am I getting enough omega-3s?

Most people in the United States get enough ALA from the foods they eat. They also get small amounts of EPA and DHA. Recommended amounts of EPA and DHA have not been established.

What happens if I don't get enough omega-3s?

A deficiency of omega-3s can cause rough, scaly skin and a red, swollen, itchy rash. Omega-3 deficiency is very rare in the United States.

What are some effects of omega-3s on health?

Scientists are studying omega-3s to understand how they affect health. People who eat fish and other seafood have a lower risk of several chronic diseases. However, it is not clear whether these health benefits come from simply eating these foods or from the omega-3s in these foods. Here are some examples of what the research has shown.

Cardiovascular disease

Many studies show that eating fatty fish and other types of seafood as part of a healthy eating pattern helps keep your heart healthy and helps protect you from some heart problems. Getting more EPA and DHA from foods or dietary supplements lowers triglyceride levels, for example.

The American Heart Association (AHA) recommends eating one to two servings of seafood per week to reduce your risk of some heart problems, especially if you consume the seafood in place of less healthy foods. For people with heart disease, the AHA recommends consuming about 1 g per day EPA plus DHA, preferably from oily fish, but supplements are an option under the guidance of a health care provider. The

AHA does not recommend omega-3 supplements for people who do not have a high risk of cardiovascular disease.

Infant health and development

During pregnancy and breastfeeding, eating 8 to 12 ounces per week of fish and other seafood may improve your baby's health. However, it is important to choose fish that are higher in EPA and DHA and lower in mercury. Examples are salmon, herring, sardines, and trout. It is not clear whether taking dietary supplements containing EPA and DHA during pregnancy or breastfeeding affects a baby's health or development. However, some studies show that taking these supplements may slightly increase a baby's weight at birth and the length of time the baby is in the womb, both of which may be beneficial. Breast milk contains DHA. Most commercial infant formulas also contain DHA.

Cancer prevention

Some studies suggest that people who get more omega-3s from foods and dietary supplements may have a lower risk of breast cancer and perhaps colorectal cancer. However, a large clinical trial found that omega-3 supplements did not reduce the overall risk of cancer or the risk of breast, prostate, or colorectal cancers. Other clinical trials in progress will help clarify whether omega-3s affect cancer risk.

Alzheimer's disease, dementia, and cognitive function

Some—but not all—research shows that people who consume more omega-3s from food such as fish may have a lower risk of developing Alzheimer's disease, dementia, and other problems with cognitive function. More study of the effects of omega-3s on the brain is needed.

Age-related macular degeneration

Age-related macular degeneration (AMD) is a major cause of vision loss among older adults. Studies suggest that people who get higher amounts of omega-3s from the foods they eat may have a lower risk of developing AMD. However, once someone has AMD, taking omega-3 supplements does not keep the disease from getting worse or slow down vision loss.

Dry eye disease

Dry eye disease occurs when tears don't provide enough moisture, causing eye discomfort and vision problems. Some studies show that getting more omega-3s from foods or supplements—mainly EPA and DHA—helps relieve symptoms of dry eye disease. However, a large, recent study found that the symptoms of people with dry eye disease who took fish oil supplements of 2,000 mg EPA plus 1,000 mg DHA daily for 1 year did not improve any more than those who took a placebo (a dummy pill). Another study in healthy people found that daily supplements of 460 mg EPA plus 380 mg DHA for 5.3 years did not affect the risk of getting dry eye disease. More research on the effects of omega-3s on dry eye disease is needed.

Rheumatoid arthritis

Rheumatoid arthritis (RA) causes chronic pain, swelling, stiffness, and loss of function in the joints. Some clinical trials have shown that taking omega-3 supplements may help manage RA when taken together with standard RA medications and other treatments. For example, people with RA who take omega-3 supplements may need less pain-relief medication, but it is not clear if the supplements reduce joint pain, swelling, or morning stiffness.

Other conditions

Researchers are studying whether taking omega-3 dietary supplements may help lessen some of the symptoms of attention-deficit/hyperactivity disorder, childhood allergies, and cystic fibrosis. However, more research is needed to fully understand the potential benefits of omega-3s for these and other conditions.

Can omega-3s be harmful?

The U.S. Food and Drug Administration recommends consuming no more than 5 g/day of EPA and DHA combined from dietary supplements. Any side effects from taking omega-3 supplements are usually mild. They include an unpleasant taste in the mouth, bad breath, heartburn, nausea, stomach discomfort, diarrhea, headache, and smelly sweat.

Do omega-3s interact with medications or other dietary supplements?

Omega-3 dietary supplements may interact with the medications you take. For example, high doses of omega-3s may cause bleeding problems when taken with warfarin (Coumadin) or other anticoagulant medicines.

Talk with your health care provider about possible interactions between omega-3 supplements and your medications.

Omega-3s and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about omega-3s?

- For general information on omega-3 fatty acids
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Omega-3 Fatty Acids](#)
 - [Omega-3 Fatty Acids](#), MedlinePlus
- For more information on food sources of omega-3 fatty acids
 - USDA's [FoodData Central](#)
 - Nutrient List for ALA (listed by [food](#) or by [ALA content](#)), USDA
 - Nutrient List for DHA (listed by [food](#) or by [DHA content](#)), USDA
 - Nutrient List for EPA (listed by [food](#) or by [EPA content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

adverse event

An unwanted medical problem that occurs during treatment. Adverse events may be unrelated to the treatment or they may be caused by the therapy or procedure. For example, an adverse event may be caused by the toxic effects of a particular drug or dietary supplement or by an interaction with another therapy. Also called adverse effect and side effect.

age-related macular degeneration

AMD. An eye disease that results in a loss of central, "straight-ahead" vision. AMD is the leading cause of vision loss in older Americans.

anticoagulant

A drug or other substance that stops blood clots from forming. Also called a blood thinner.

association

A relationship between two conditions or states such that if one is present, the other is likely to be present as well. An association between two conditions or states, however, does not necessarily imply a cause and effect relationship. The terms association and relationship are often used interchangeably.

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

breast cancer

Uncontrolled growth of abnormal cells in the breast.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

chronic

Happening for a long time, persistently, or repeatedly.

chronic disease

A condition that is continuous or recurrent, is not easily cured, and cannot be passed from person to person. Examples of chronic diseases include heart disease, diabetes, and asthma.

clinical trial

A type of research study that uses volunteers to test the safety and efficacy (the ability to produce a beneficial effect) of new methods of screening (checking for disease when there are no symptoms), prevention, diagnosis, or treatment of a disease. Also called a clinical study.

cognitive function

Mental awareness and judgment.

colorectal cancer

Uncontrolled growth of abnormal cells in the colon (the longest part of the large intestine) and/or the rectum (the last several inches of the large intestine before the anus).

consume

To eat or drink.

cystic fibrosis

A common inherited disease that causes the body to make thick, sticky mucus that builds up in the lungs and blocks the airways, leading to repeated serious lung infections. Mucus also blocks the pancreas, which stops digestive enzymes from reaching the intestines. Cystic fibrosis also causes very salty sweat, which can lead to dehydration, increased heart rate, tiredness, low blood pressure, and heat stroke.

deficiency

An amount that is not enough; a shortage.

dementia

Damaged brain function (thinking, learning, making decisions, remembering) that worsens over time. It disrupts activities of daily living such as bathing, dressing, and walking.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

endocrine system

Hormone-producing glands that affect growth and development, hunger, metabolism (chemical changes in the body), sleep, sexual function, and mood. The endocrine system includes the thyroid gland, pituitary gland, adrenal gland, parathyroid glands, adrenal glands, pancreas, ovaries, and testicles.

essential

In nutrition, essential nutrients are ones that we must consume for good health because our bodies cannot make them. Essential nutrients include vitamins and minerals.

Food and Drug Administration

FDA, Department of Health and Human Services. FDA is the Federal government agency responsible for ensuring that foods and dietary supplements are safe, wholesome and sanitary, and that drugs, medical devices, cosmetics, and food are honestly, accurately and informatively represented to the public. FDA regulates dietary supplements under a different set of regulations than those covering conventional foods and drug products (prescription and over-the-counter). The dietary supplement manufacturer is responsible for ensuring that a dietary supplement is safe before it is marketed. FDA is responsible for taking action against any unsafe dietary supplement product after it reaches the market. Generally, manufacturers do not need to get FDA approval before producing or selling dietary supplements.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gland

A small organ that makes and releases a substance such as sweat, tears, saliva, milk, a hormone, or substances that aid in digestion.

gram

g. A measure of weight. It is a metric unit of mass equal to 0.001 kilogram (it weighs 0.035 ounce).

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant formula

An artificial form of breast milk.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

omega-3 fatty acid

A main component of fats used by the body for energy and tissue growth. Omega-3s are essential fatty acids in the human diet; they are found in fish oil and certain plant and nut oils.

placebo

An inactive substance or treatment that has no effect on the body and that ideally looks, smells, and tastes the same as, and is given the same way as, the active drug or treatment being tested. The effects of the active substance or treatment are compared to the effects of the placebo.

retina

The light-sensitive layers of nerve tissue at the back of the eye that receive images and send them as electric signals through the optic nerve to the brain.

rheumatoid arthritis

An autoimmune disease that causes pain, swelling, and stiffness in the joints. It may cause severe joint damage, loss of function, and disability. The disease may last from months to a lifetime, and symptoms may improve and worsen over time.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

triglyceride

A type of fat found in your blood. When you eat more than you need, your body turns the excess calories into triglycerides. High blood levels of triglycerides can increase your risk of heart disease, heart attack, and stroke.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vegetarianism

The practice of avoiding all or most animal products for environmental, philosophical, and health reasons. Vegetarians (people who practice vegetarianism) eat a diet based on foods that come from plants and may include some dairy products and eggs. See: vegetarian diet.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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